01 / PROBLEM AND CONTRIBUTION

Active-load common-source amplifier

EEE 433/591 · Analog IC Design · Lab 3 | Fall 2025

RESULT AT A GLANCE

Designed an active-PMOS-load common-source stage with reported simulated gain of 31.33 dB and 1.103 MHz bandwidth, then compared it with a cascoded implementation.

Engineering question

How does an active load increase gain, and what changes when the signal path is cascoded?

Situation

An active PMOS load supplies a high small-signal resistance without a large physical resistor. The report compares a simple common-source stage with a cascoded circuit and examines the resulting frequency response and distortion.

Task

Bias the active-load stage, compare hand analysis with simulation, and investigate gain, bandwidth, transfer characteristics, and clipping.

My contribution

Prepared operating-point schematics, AC/transient simulations, spectrum plots, and analytical comparisons for the two reported amplifier configurations.

Tools and concepts

Analog IC design / Active loads / Cadence Virtuoso / Frequency response / Distortion analysis

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation

Established bias and device sizing for the 2.5 V, 1 pF-load study. Compared the simple-stage simulated branch current of 51.29 µA with the 50 µA hand calculation.

02 / Implementation

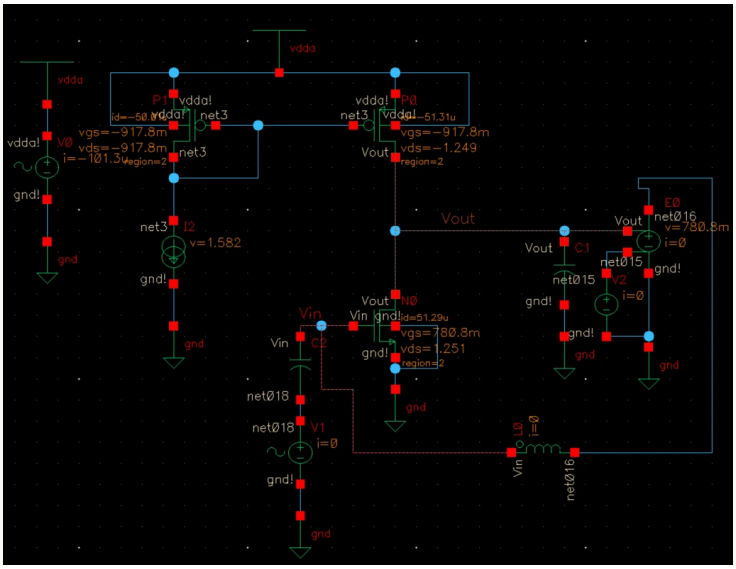
Used operating-point, AC, and DC-transfer plots to evaluate the active-load stage. The simple-stage table reports 31.33 dB gain and 1.103 MHz bandwidth, compared with 33.98 dB and 1.59 MHz in the hand analysis.

03 / Implementation

Increased input amplitude and inspected output spectra to explain clipping-related harmonics. Compared the simple stage with a cascoded implementation while retaining the report's conflicting summary values as a documentation limitation.

Bias before bandwidth

The operating point establishes transconductance and output resistance. AC analysis linearizes around that point, while transient amplitude sweeps reveal where the linear approximation breaks down.



Active-load amplifier operating point

Source report p. 3: transistor operating-point annotations.

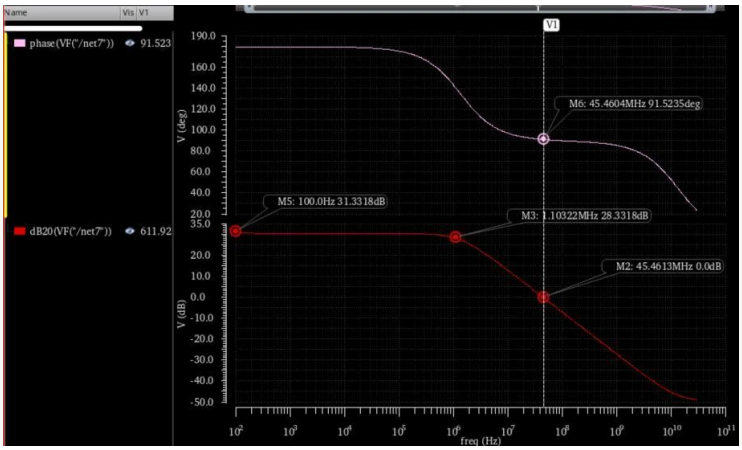
03 / RESULTS AND PERSPECTIVE

Evidence and engineering perspective

Simulated gain	31.33 dB
-3 dB bandwidth	1.103 MHz
Simulated branch current	51.29 µA

Reported result

The simple common-source stage reports 31.33 dB gain, 1.103 MHz bandwidth, and 51.29 µA branch current. The report also demonstrates a higher-gain cascoded variant, but its conflicting summary tables prevent an unqualified numerical comparison.



Simple-stage gain and phase

Source report p. 4: gain and bandwidth markers for the simple common-source stage.

Scope and limitations

The cascode section lists inconsistent bandwidth and gain summaries, so its numbers are not used as headline results. Output peak readings are not treated as peak-to-peak linear swing, and phase at unity gain is not relabelled as phase margin.

Possible next steps - proposed work

- Reconcile the cascoded-stage tables against the original AC markers.
- Measure harmonic distortion over an explicitly stated amplitude sweep.

Source material: EEE 433/591 Lab 3 submitted report, physical PDF pp. 3-8, 11-18; simple-stage result table on p. 8.